RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 7 October 2026 – NumPy Numerical Processing

2. DAILY LOG ENTRY FORM

- **Date:** 07 October 2026

- **1. Assigned Task:**

  NumPy array operations and vectorized numerical processing routines.

- **2. What Was Studied:**

  Studied NumPy arrays in detail, including array creation, dimensions, shape, indexing, matrix structure, element-wise numerical operations, and vectorized processing. Also understood how these operations can be applied to structured numerical telemetry data for faster and cleaner data processing.

- **3. What Was Implemented:**

  Created a numerical telemetry dataset and converted it into a NumPy matrix. Performed array creation, shape and dimension checks, indexing, basic numerical calculations, and vectorized operations on the dataset. Verified the final telemetry matrix shape and checked that the expected outputs were produced.

- **4. Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.10 , Python .venv , NumPy

- **5. Output File Name-** suraj_rudra_task/Day3/notes/numpy_numerical_Processing

  Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**

  No major problem was encountered during NumPy numerical processing.

- **7. How Problem Was Resolved:**

  No major problem has aquired

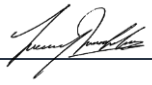
- **8. Next Task Scheduled:**

  (08 October 2026)-Pandas-based telemetry ingestion,time-indexing,and query handling

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 07/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026